Topic 9: Thermodynamics

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the relationship between the volume and temperature of a fixed mass of gas.

Mathematical skills that could be developed in this topic include substituting numerical values into algebraic equations using appropriate units for physical quantities.

This topic may be studied using applications that relate to thermodynamics, for example space technology.

Students should:
144. be able to use the equations $\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$
145. **CORE PRACTICAL 12: Calibrate a thermistor in a potential divider circuit as a thermostat.**
146. **CORE PRACTICAL 13: Determine the specific latent heat of a phase change.**
147. understand the concept of *internal energy* as the random distribution of potential and kinetic energy amongst molecules
148. understand the concept of *absolute zero* and how the average kinetic energy of molecules is related to the absolute temperature
149. be able to derive and use the equation $pV = \frac{1}{3}Nm\langle c^2\rangle$ using the kinetic theory model
150. be able to use the equation $pV = NkT$ for an ideal gas
151. **CORE PRACTICAL 14: Investigate the relationship between pressure and volume of a gas at fixed temperature.**
152. be able to derive and use the equation $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$
153. understand what is meant by a *black body radiator* and be able to interpret radiation curves for such a radiator
154. be able to use the Stefan-Boltzmann law equation $L = \sigma AT^4$ for black body radiators
155. be able to use Wien's law equation $\lambda_{max}T$ = 2.898 x 10^{-3} m K for black body radiators.